

Kisan Vigyaan (CropShield AI): Project Synthesis

Fieldwork Documentation & Farmer Testimonials

Regions: Thanjavur, Vellore, Kallakurichi, Namakkal

1. Thanjavur (Orathanadu Block) - Arumugam's Account

Crop: Paddy (Rice) | **Threat:** Brown Plant Hopper (BPH)

The canals were full during the last Samba season, and my paddy fields were waterlogged. I had planted the crops close together and used a lot of nitrogen fertilizer to improve the yield. The result was a very thick and humid crop around the base of the plants.

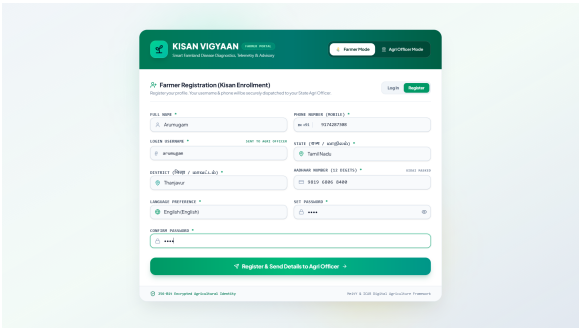
After some time, I noticed small patches of the field turning golden-brown. It turned out to be a Brown Plant Hopper attack. Large numbers of hoppers had gathered near the base of the plants and were feeding on the sap.

Earlier, we would panic and spray chemicals across the field, even though the spray often did not reach the base of the plants where the hoppers were. This time, the Kisan Vigyaan system detected the high humidity through field sensors and identified the problem using satellite data. The app gave instructions in Tamil. I drained the standing water and cleared narrow paths through the field so that air could move better. The lower humidity helped control the pest and the crop recovered.

Fieldwork photograph



Registration screenshot - matching username



2. Vellore (Kaniyambadi Block) - Karthik's Account

Crop: Groundnut (Peanut) | Threat: Tikka Leaf Spot & Wild Boars

It has been a difficult year for us in Kaniyambadi, Vellore. The red soil became very dry and hard, and we depended heavily on deep tube wells. After the hot weather returned, dark brown spots with yellow rings appeared on the groundnut leaves. The lower leaves started falling off and the plants became weak. It was a serious Tikka Leaf Spot infection.

We often mistake these spots for a nutrient problem and add more urea, which can make the fungal infection worse. Wild boars were also entering the fields at night, digging through the hard soil and damaging the developing groundnut pods.

The soil probes helped us decide when to run drip irrigation so the soil stayed soft enough for the groundnut pegs to develop. I uploaded a photo of an infected leaf, and the system identified it as Tikka Spot and suggested the required dose of Mancozeb fungicide. For the wild boars, we used the smart acoustic scare system along the field boundary. When movement was detected at night, it played sounds such as dogs barking and firecrackers. The boars moved away, so we did not have to use electric fencing.

Fieldwork photograph



Registration screenshot - matching username

A screenshot of the 'KISAN VIGYAN' Farmer Registration (Kisan Enrollment) form. The form is titled 'Farmer Registration (Kisan Enrollment)' and includes a 'Login' button and a 'Register' button. The form fields are organized into two columns. The left column contains: 'Full Name' (with a dropdown arrow), 'Village' (with a dropdown arrow), 'District' (with a dropdown arrow), 'Mobile Number' (with a dropdown arrow), 'Email' (with a dropdown arrow), and 'Password' (with a dropdown arrow). The right column contains: 'PAN Number' (with a dropdown arrow), 'Aadhaar Number' (with a dropdown arrow), 'Land Record Number' (with a dropdown arrow), 'Caste' (with a dropdown arrow), 'Gender' (with a dropdown arrow), and 'Date of Birth' (with a dropdown arrow). At the bottom of the form, there is a green button labeled 'Register & Send Details to AgrOfficial' and a small text link 'Go to Kisan Vigyan App'.

3. Kallakurichi (Chinnasalem Block) - Selvam's Account

Crop: Hybrid Maize (Corn) | Threat: Fall Armyworm (FAW)

Farming in the dry areas of Chinnasalem, Kallakurichi is always difficult because water is limited. We planted hybrid maize in our black cotton soil after a long dry spell, hoping for a good harvest. Within a few days, the young plants were badly damaged. The central leaves had large holes and insect droppings, and Fall Armyworm caterpillars had moved into the centre of the plants and were feeding on the new growth.

Usually, we react only after the worms appear. We buy strong chemical mixtures and apply them directly to the plants. It costs more, and sometimes the plant has already been damaged by the time we treat it.

This year, the system helped us act earlier. The automated misting nozzles were set to spray a natural mixture of neem oil and *Vitex negundo* twice a day, at dawn and dusk. This helped reduce the chance of moths laying eggs on the plants. The satellite moisture data also showed which parts of the field were under drought stress, so we gave those areas additional water.

Fieldwork photograph



Registration screenshot - matching username



KISAN VIGYAN
Digitize your farm, Grow your income

[Home Page](#)
[Agri-Office/Shop](#)

Farmer Registration (Kisan Enrollment)

Register your profile. You can activate your online using digital Sign by your State Agri-Office

[Login](#)
[Register](#)

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4. Namakkal (Agasteeswaram Block) - Arun's Account

Crop: Coconut & Banana Plantations | Threat: Rhinoceros Beetle

Down here in Agasteeswaram, Namakkal, we receive heavy rainfall from both monsoons because we are close to the Western Ghats. The weather is good for coconut and Nendran banana plantations, but the high moisture also creates pest problems.

After several weeks of heavy rain, the organic manure pits became good places for pests to breed. Soon, I noticed V-shaped cuts appearing on the leaves of my three-year-old coconut palms. The damage was caused by Rhinoceros Beetles, which had entered the crowns and fed on the young buds.

The damage is difficult to notice because it happens high inside the crown. By the time the damaged leaves unfold, the beetles may have already moved on. Earlier, we had to climb the wet trees to remove them or pack the leaf joints with sand and naphthalene balls.

I uploaded a photo of the damaged leaves, and the Kisan Vigyaan platform identified the pattern as a possible Rhinoceros Beetle attack and provided an audio guide in Tamil. The automated trunk-line micro-misters were then used to release camphor and neem mist into the crown area during the evening, when the beetles are active. This helped protect the young buds without requiring anyone to climb the trees.

Fieldwork photograph



Registration screenshot - matching username

A screenshot of the Kisan Vigyaan Farmer Registration (Kisan Enrollment) form. The form is titled "Farmer Registration (Kisan Enrollment)" and includes a "Login" button. The form fields are: Full Name (Arun), Phone Number (9876543210), Email Address (arun@kisanvignyaan.com), District (Namakkal), and State (Tamil Nadu). There are also fields for "Crop" and "Threat". The form is submitted by clicking the "Register & Save Details to Agr. Officer" button. At the bottom, there is a note: "We Will Send you a SMS on your registered mobile number".